

REAL ESTATE HOUSING MARKET ANALYSIS

An End-to-End Data Analytics Project

100,000 Total Transactions	DKK 193Bn Total Market Value	DKK 1.93M Avg Property Price	1992 – 2024 Analysis Period	50+ Cities Covered
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Tools Used:

Python (Pandas, Matplotlib, Seaborn) • MySQL (SQL Analysis) • Power BI (Dashboard)

Data Analytics Portfolio Project | 2026

1. Business Problem Statement

A real estate consulting and investment firm seeks to gain deeper insights into housing market dynamics to support data-driven investment, pricing, and market expansion decisions. The firm has observed significant variations in property prices, sales activity, and buyer negotiation behavior across different cities and regions. Changing economic conditions — particularly inflation and interest rates — are also suspected to influence property demand and valuation.

The overarching business question driving this project is:

“How can real estate transaction and economic data be leveraged to identify market trends, evaluate investment opportunities, optimize pricing strategies, and support data-driven decision-making in the housing market?”

1.1 Key Objectives

- Identify cities and regions with the highest property values and strongest market performance.
- Understand the impact of property characteristics (house type, size, rooms, build era) on purchase prices.
- Analyze how inflation and interest rates influence housing prices and market activity.
- Evaluate buyer negotiation behavior across different regions and property segments.
- Discover emerging markets and investment opportunities through geographic and trend analysis.
- Develop an interactive analytics dashboard to support strategic business and investment decisions.

2. Dataset Overview

The dataset contains 100,000 residential property transactions from Denmark, spanning 1992 to 2024. It includes property-level details, geographic identifiers, macroeconomic indicators, and engineered buyer behavior metrics. The data was sourced for academic and portfolio purposes.

Data columns (total 19 columns):				
#	Column	Non-Null Count		Dtype
0	date	100000	non-null	object
1	quarter	100000	non-null	object
2	house_id	100000	non-null	int64
3	house_type	100000	non-null	object
4	sales_type	100000	non-null	object
5	year_build	100000	non-null	int64
6	purchase_price	100000	non-null	int64
7	%_change_between_offer_and_purchase	100000	non-null	int64
8	no_rooms	100000	non-null	int64
9	sqm	100000	non-null	int64
10	sqm_price	100000	non-null	float64
11	address	100000	non-null	object
12	zip_code	100000	non-null	int64
13	city	99989	non-null	object
14	area	100000	non-null	object
15	region	100000	non-null	object
16	nom_interest_rate%	100000	non-null	float64
17	dk_ann_infl_rate%	99923	non-null	float64
18	yield_on_mortgage_credit_bonds%	99923	non-null	float64

Figure 1: Dataset schema from df.info() — 19 original columns, 100,000 rows with some missing values in city, inflation rate, and mortgage yield fields

	date	quarter	house_id	house_type	sales_type	year_build	purchase_price	%_change_between_offer_and_purchase	no_rooms	sqm	sqm_price	address
0	10/24/2024	2024Q4	16	Villa	regular_sale	1997	6500000	-3	5	142	45774.650	Spættevej 16
1	10/24/2024	2024Q4	13	Apartment	regular_sale	1885	3400000	0	2	46	73913.050	Jakob Dannefærds Vej 10B, 4. th
2	10/23/2024	2024Q4	60	Villa	regular_sale	1949	4550000	-4	4	112	40625.000	Vestre Plantagevej 5
3	10/23/2024	2024Q4	29	Villa	regular_sale	2001	1630000	-12	4	186	8763.440	Toftebjerg 5
4	10/22/2024	2024Q4	92	Apartment	regular_sale	1965	1975000	-10	3	82	24085.365	Ryttergårdsve 20, 1. 11E

Figure 2: Sample rows from the raw dataset showing property transactions with all key columns

2.1 Column Description

Column	Type	Description
date	Date	Transaction date
quarter	String	Quarter label (e.g., 2024Q1)
house_id	Integer	Unique property identifier
house_type	String	Villa, Apartment, Townhouse, Summerhouse, Farm
sales_type	String	Regular sale, family sale, auction, other
year_build	Integer	Year the property was constructed
purchase_price	Integer	Final transaction price (DKK)
%_change_between_offer_and_purchase	Integer	Percentage change from offer to final purchase price
no_rooms	Integer	Number of rooms in the property
sqm	Integer	Property size in square meters
sqm_price	Float	Price per square meter (DKK)
address / zip_code / city / area / region	String	Geographic identifiers at multiple levels
nom_interest_rate%	Float	Nominal interest rate at time of sale
dk_ann_infl_rate%	Float	Danish annual inflation rate
yield_on_mortgage_credit_bonds%	Float	Mortgage bond yield — proxy for borrowing cost

2.2 Engineered Features

The following columns were derived during feature engineering to enrich the analytical depth of the dataset:

Feature	Logic / Purpose
price_segment	Classifies properties into Budget, Mid, Premium, Luxury, Ultra-Luxury based on purchase price thresholds
build_era	Categorizes year_build into: Pre-1900, 1900s, Mid-century, 70s, 80-90s, 2000s, Modern
sales_season	Derives Winter, Spring, Summer, Autumn from transaction date
offer_price	Back-calculated from purchase_price and %_change column to reveal initial listing price
negotiation_gap	Absolute DKK difference between offer price and purchase price
real_interest_rate	nom_interest_rate% minus dk_ann_infl_rate% — measures true cost of borrowing
distressed_sale	Flag where %_change_between_offer_and_purchase < -15% (severe price cut)

3. Data Cleaning & Preparation (Python)

All data cleaning and preprocessing was performed using Python (Pandas library). The following steps were executed to ensure data quality before analysis.

3.1 Initial Assessment

- Loaded dataset using `pd.read_csv()` and reviewed structure with `df.head()`, `df.info()`, and `df.describe()`.
- Identified date format inconsistency: dates existed in both 'MM/DD/YY' and 'MM-DD-YY' formats.
- Found missing values in three columns: city (11 nulls), `dk_ann_infl_rate%` (77 nulls), and `yield_on_mortgage_credit_bonds%` (77 nulls).

3.2 Cleaning Steps

- Date normalization: Standardized all date entries to DD/MM/YY format using regex-based parsing and `pd.to_datetime()`.
- City nulls: Rows with missing city values were dropped as city is a critical geographic dimension and represents less than 0.02% of total data.
- Inflation & yield nulls: Missing values in `dk_ann_infl_rate%` and `yield_on_mortgage_credit_bonds%` were imputed with the median of the same quarter within the same year — preserving seasonal economic patterns.

3.3 Export to MySQL

After cleaning and feature engineering, the final dataset was exported to a MySQL server using SQLAlchemy for structured SQL analysis. This allowed seamless transition from Python-based EDA to SQL-driven business querying.

4. Exploratory Data Analysis (Python)

EDA was conducted using Pandas, Matplotlib, and Seaborn to uncover distributions, outliers, patterns, and relationships across property and economic dimensions.

4.1 Property Price Distribution

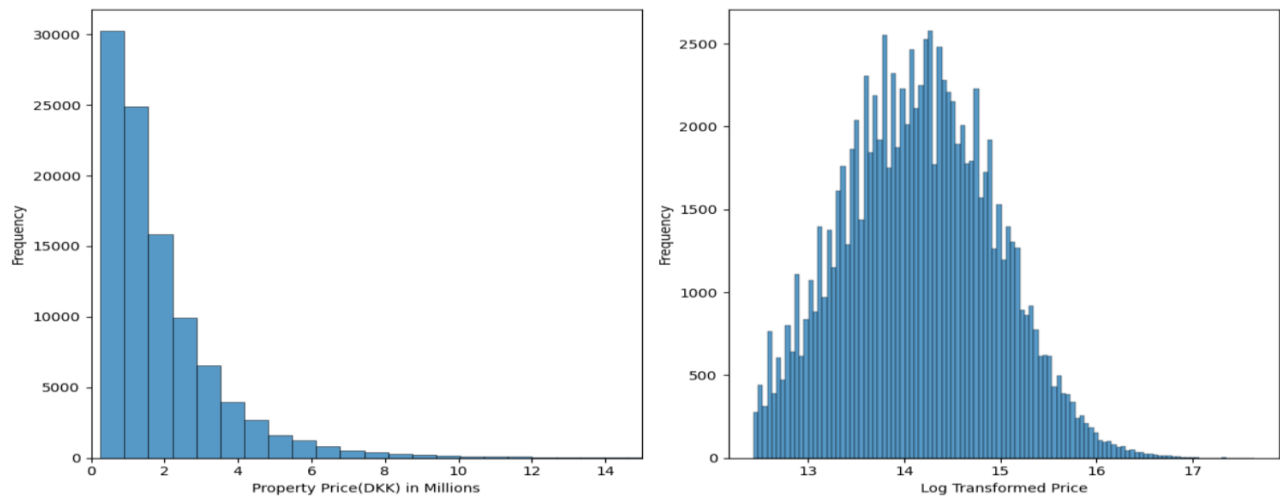


Figure 3: Property price distribution (left: raw DKK, right: log-transformed) — strong right skew indicates most properties are modestly priced with a small segment of luxury outliers

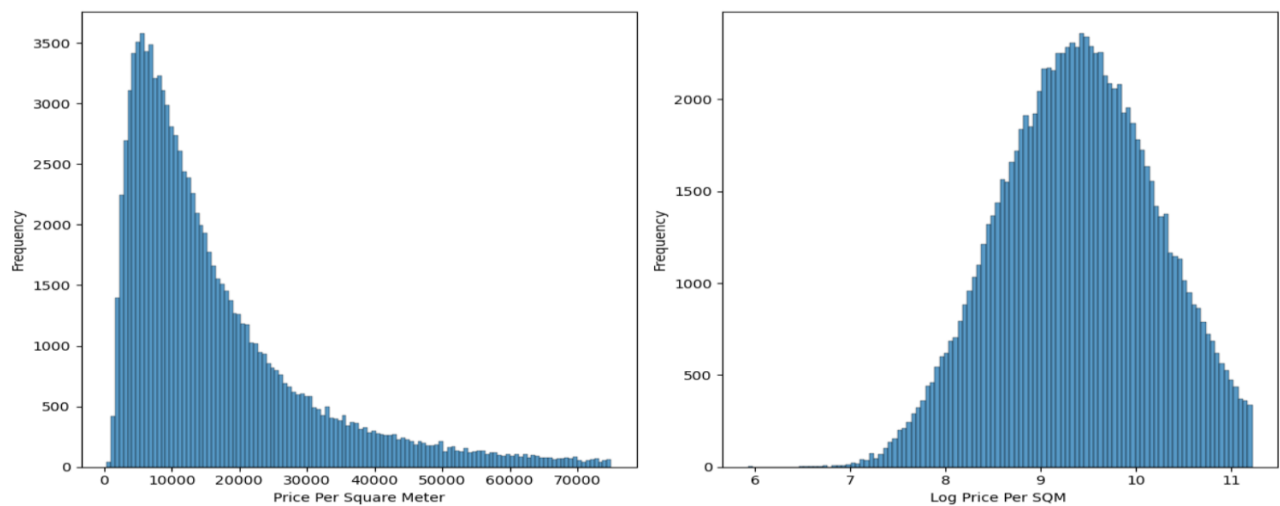


Figure 4: Price per square meter distribution (left: raw, right: log-transformed) — log-normal distribution confirms SQM pricing follows a multiplicative structure

□ Price Distribution Findings

- Most properties are modestly priced; a small luxury segment pulls average prices significantly above medians.

- The log-normal distribution of both purchase price and SQM price validates the use of median (not mean) as the representative price metric.
- Price per SQM shows less extreme variation than total price, making it a more reliable cross-property comparison metric.

4.2 House Type Analysis

House Type	Count	Share	Mean Price	Median Price	M/Md Ratio
Villa	54,210	54.2%	1,785,136	1,300,000	1.37
Apartment	19,354	19.4%	2,433,244	1,950,000	1.25
Summerhouse	11,339	11.3%	1,219,568	977,500	1.25
Townhouse	10,184	10.2%	2,107,443	1,720,000	1.23
Farm	4,902	4.9%	2,740,145	1,700,000	1.61

House Type	Median SQM (m²)	Median SQM Price	Max Price	Min Price	
Apartment	79	25,437	34,402,208	250,750	
Farm	186	9,415	37,250,000	250,536	
Summerhouse	76	12,944	45,955,000	250,750	
Townhouse	107	16,526	18,750,000	252,000	
Villa	147	9,109	34,000,000	250,200	

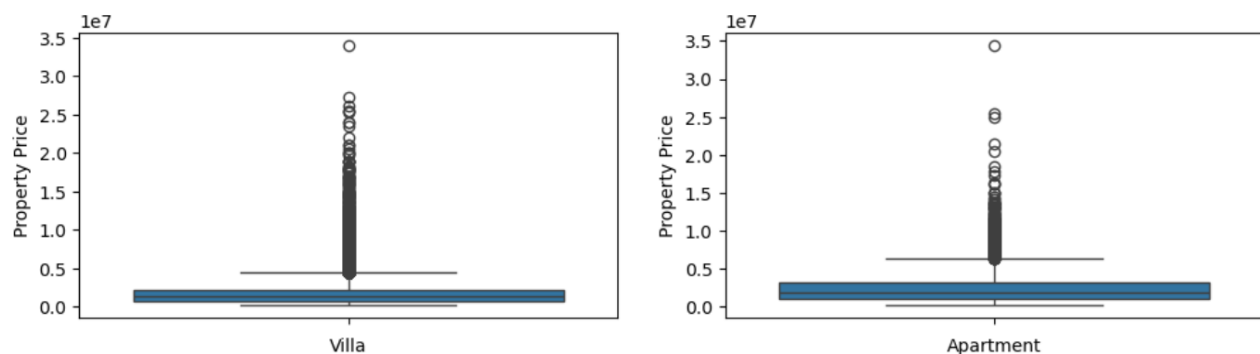


Figure 5: Boxplots — Villa and Apartment price distributions, showing presence of high-value outliers in both segments

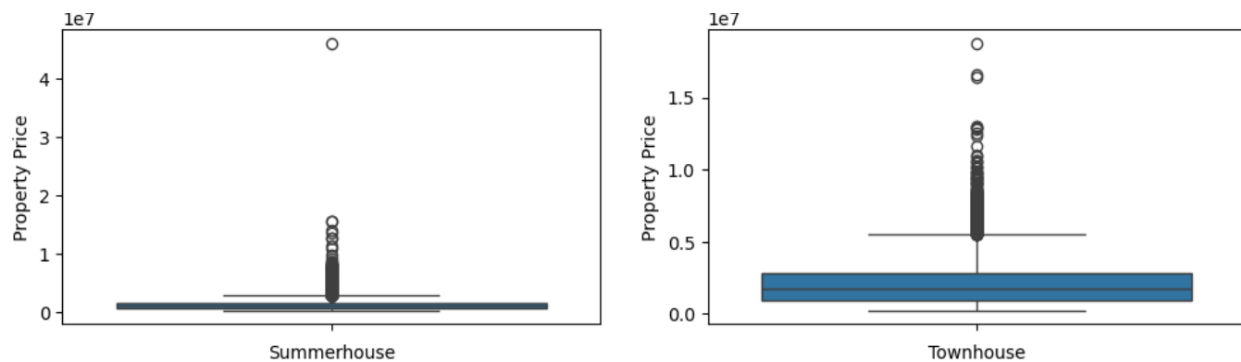


Figure 6: Boxplots — Summerhouse and Townhouse, with Summerhouse showing an extreme outlier at ~DKK 46M (likely coastal premium property)

House Type Key Insights

- Villas dominate the market (54.2% of all transactions), making them the benchmark property type.
- Apartments command the highest SQM price (DKK 25,437/m²) despite being the smallest — reflecting prime urban location premiums.
- Farms have the widest Mean/Median ratio (1.61), indicating a highly skewed market with a few extremely expensive agricultural estates.
- Townhouses show the most consistent pricing (lowest M/Md ratio at 1.23) — suitable for stable investment.
- Summerhouses have the highest recorded transaction (DKK 45.9M), likely coastal luxury vacation properties.

4.3 Regional Analysis

Region	Apartments	Farms	Summerhouses	Townhouses	Villas	Distress%
Zealand	10,139	1,089	4,910	4,926	18,676	2.10%
Jutland	7,864	3,224	5,554	3,876	29,408	2.90%
Fyn & Islands	1,300	529	660	1,192	5,583	3.65%
Bornholm	51	60	215	190	543	3.97%

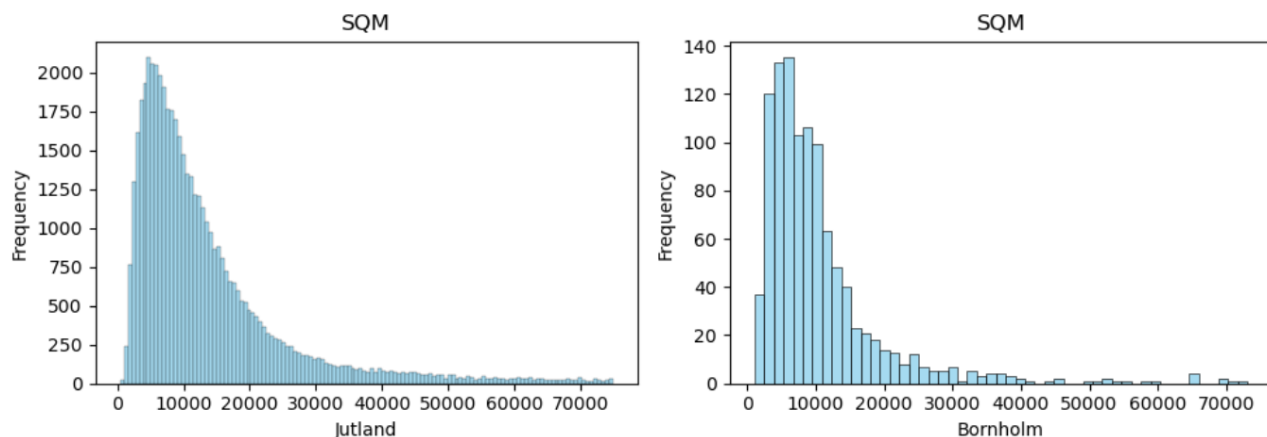


Figure 7: SQM price distribution for Jutland (left) and Bornholm (right) — both right-skewed, Bornholm shows higher SQM concentration at lower values

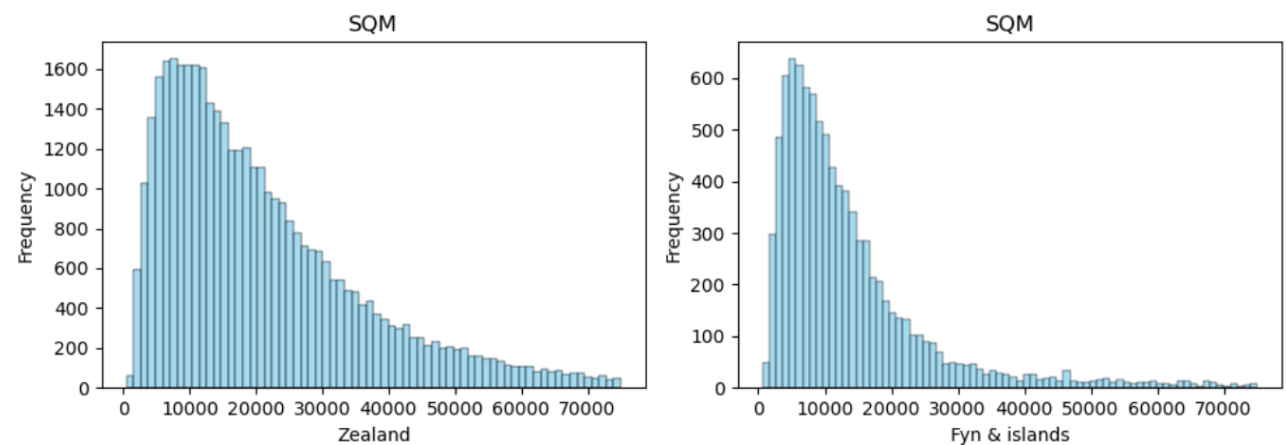


Figure 8: SQM price distribution for Zealand (left) and Fyn & Islands (right) — Zealand shows a broader, higher-value distribution indicating premium pricing

Regional Performance Insights

- Zealand is the largest and most premium market (40K transactions, IQR spread DKK 2.1M) with the lowest distressed sale rate (2.10%) — indicating strong demand and market health.
- Jutland is the most active region overall (50K transactions) but shows more price variability with a moderate distress rate of 2.90%.
- Bornholm has the smallest market but the highest distress rate (3.97%), suggesting weaker demand and lower liquidity — higher-risk investment territory.
- Fyn & Islands is a mid-tier market with solid volume but higher distress than the top two regions.

4.4 Build Era Analysis

Build Era	Median SQM Price (DKK)	Count
Pre-1900	10,154	11,535
1900s	13,014	20,444
Mid-century (1940–60s)	11,261	26,707
70s	10,986	22,950
80-90s	13,987	3,393
2000s	15,950	7,335
Modern (Post-2010)	17,348	7,625

Build Era Insights

- Modern properties (post-2010) command the highest premium at DKK 17,348/m² — reflecting energy efficiency, modern amenities, and current building standards.

- The 2000s and Modern eras together represent a premium segment despite lower absolute transaction counts.
- Mid-century (1940–60s) and 70s builds have lower SQM prices, likely requiring renovation investment.
- Surprisingly, 1900s properties have higher SQM values than pre-1900 — possibly due to historical/character value in prime city locations.

4.5 Negotiation Behavior Analysis

House Type	Zero Neg. %	Mean Gap (DKK)		Interpretation
Farm	80.7%	-237,182		Rare but deep discounts
Apartment	69.1%	-104,608		Urban demand keeps sellers firm
Townhouse	66.4%	-100,377		Moderate negotiation activity
Villa	61.1%	-125,792		Active negotiation, large gaps
Summerhouse	59.6%	-94,530		Most negotiated, smallest gaps

□ Negotiation Behavior Insights

- 63.96% of all transactions (63,957 out of 99,989) have zero negotiation gap — sellers and buyers agree on the offer price outright.
- Farms have the lowest negotiation frequency (80.7% zero-gap) but the largest average discount when negotiation occurs (DKK -237,182).
- Summerhouse buyers are the most active negotiators (40.4% of transactions have a gap), though discount amounts are the smallest.
- The extreme left skew (skewness = -16.1) confirms rare but massive discounts drag the mean far below zero.
- Villas, despite dominating the market, see meaningful negotiation in ~39% of cases with substantial discounts averaging DKK -125,792.

5. Data Analysis Using SQL (MySQL)

Eighteen targeted business questions were answered using structured SQL queries on the cleaned property sales dataset loaded into MySQL. Queries covered pricing trends, regional comparisons, interest rate impact, negotiation behavior, and investment screening.

5.1 Geographic & Pricing Analysis

Q1 — Top 10 Cities by Average Purchase Price

The query retrieved the 10 highest-valued cities by mean purchase price. The Power BI dashboard confirms Rungsted Kyst leads at DKK 6.1M average, followed by Nordhavn (6.0M), Klampenborg (6.0M), and Vedbæk (5.8M). All top cities are concentrated in the affluent North Zealand coastal area near Copenhagen.

Q2 — Average SQM Price by Region

Zealand commands the highest average SQM price among all regions, consistent with its role as the premium market. Jutland leads in raw volume but trails in per-square-meter valuation. The query confirms geographic price hierarchy across Denmark's four regions.

5.2 Temporal Trend Analysis

Q3 — Average Purchase Price by Quarter & Year

Year	Quarter	avg_price
2024	Q1	2620185.8451
2024	Q2	2656255.7373
2024	Q3	2406010.4905
2024	Q4	2507224.0260
2023	Q1	2522719.9160
2023	Q2	2487445.7788
2023	Q3	2527115.6007
2023	Q4	2519395.0639
2022	Q1	2675487.1208
2022	Q2	2548998.1705
2022	Q3	2345789.6548
2022	Q4	2362730.8761
2021	Q1	2617485.1993
2021	Q2	2476720.7069
2021	Q3	2535178.0443
2021	Q4	2623165.9137
2020	Q1	2376367.9386
2020	Q2	2237373.3810
2020	Q3	2429282.9606
2020	Q4	2507545.5803
2019	Q1	2117920.3456
2019	Q2	2232878.9852
2019	Q3	2281112.8071
2019	Q4	2239440.8527
2018	Q1	2167860.2149

Figure 9: Quarterly average purchase price trend (partial view) — prices in recent years (2019–2024) range DKK 2.1M to 2.7M with consistent seasonal patterns

The data shows consistent upward price movement over the analysis period. Recent years (2020–2024) show average prices between DKK 2.1M and DKK 2.7M. Q1 2022 stands as one of the highest averages at DKK 2.68M, before moderating in 2023–24 as interest rates rose sharply.

Q4 — Highest Sales Volume by Quarter

Year	Quarter	sale_volume	total_sale
2021	Q1	5923369006	2263
2021	Q2	5560237987	2245
2020	Q4	4869653517	1942
2020	Q3	4501461326	1853
2022	Q1	4452010569	1664
2021	Q3	4411209797	1740
2023	Q4	4217467337	1674
2021	Q4	4165587471	1588
2024	Q2	4135790183	1557
2022	Q2	3945849168	1548
2020	Q2	3852756962	1722
2023	Q2	3666495078	1474
2019	Q3	3512913723	1540
2019	Q2	3481058338	1559
2024	Q1	3314535094	1265
2019	Q4	3177766570	1419
2020	Q1	3134429311	1319
2023	Q1	3062581978	1214
2018	Q2	3055928726	1417

Figure 10: Sale volume by quarter — 2021 Q1 and Q2 show peak volumes, coinciding with the post-pandemic property boom in Denmark

□ Temporal Trend Insights

- 2021 Q1 recorded the highest single-quarter sale volume (2,263 transactions, DKK 5.9Bn) — driven by post-pandemic demand surge and historically low interest rates.
- Prices showed steady growth from 1992 through 2022, with the most aggressive appreciation between 2010 and 2022.
- 2023 and 2024 saw moderation in both price growth and volume, consistent with rising real interest rates in this period.

5.3 Property Characteristics Analysis

Q5 — House Type with Highest Average Purchase Price

Farms rank as the highest-priced property type by average purchase price (DKK 2.74M), despite having the lowest SQM price. This reflects large land parcels and the overall size of farm estates. However, the mean is skewed by a small number of exceptionally expensive agricultural properties.

Q6 — Number of Rooms vs. Average Price

A near-linear positive relationship exists between room count and average purchase price, as confirmed by the Power BI chart. Properties with 15+ rooms reach average prices above DKK 6M, while 1-2 room properties average around DKK 1.5–1.8M. This validates room count as a reliable price predictor.

Q7 & Q9 — Build Era and SQM Price

build_era	house_type	avg_sqm_price
1900s	Apartment	29198.3822666073
1900s	Farm	13563.411999421634
1900s	Summerhouse	19273.03466508264
1900s	Townhouse	17372.281884478696
1900s	Villa	12835.296051114046
2000s	Apartment	31425.246156119443
2000s	Farm	12209.767541139248
2000s	Summerhouse	16348.54838579135
2000s	Townhouse	21889.48124637554
2000s	Villa	12794.945663204138
70s	Apartment	21728.86766284101
70s	Farm	15484.90063833876
70s	Summerhouse	14324.49514535129
70s	Townhouse	15780.220867139318
70s	Villa	11357.512557957152
80-90s	Apartment	31178.33791733333
80-90s	Farm	16425.076218072285
80-90s	Summerhouse	16158.546141843108
80-90s	Townhouse	19812.495772706934
80-90s	Villa	13655.27639193461
mid_century	Apartment	23637.614723375515
mid_century	Farm	14066.28360413793
mid_century	Summerhouse	15120.413096734073
mid_century	Townhouse	20657.890529286826
mid_century	Villa	12385.90331011282

Figure 11: Average SQM price by build era and house type — Apartments consistently command higher SQM prices across all eras; 2000s and Modern eras show premiums

□ Build Era & Type Cross-Analysis Insights

- Across every build era, Apartments consistently hold the highest SQM price — peaking at DKK 31,425/m² for 2000s-built apartments.
- Modern-era Villas (DKK ~15,000/m²) and Townhouses (DKK ~20,000/m²) are the best-performing non-apartment segments.
- Farm SQM prices remain low across all eras (DKK 9,000–16,000/m²), reinforcing that farm value lies in land area, not SQM price.

- 80-90s builds show a value dip across types — suggesting structural or maintenance issues that buyers discount for.

Q8 — City with Largest Villas

SQL analysis identifies the city with the highest average SQM for villas, highlighting which markets have the largest villa properties — useful for investors targeting premium-size villa segments.

5.4 Economic & Macro Analysis

Q10 — Nominal Interest Rate vs. Average Purchase Price

Year	Quarter	Nominal_interest_rate	avg_price
1992	Q1	9.5	857615.2756
1992	Q2	9.5	889696.8803
1992	Q3	9.5	826899.8207
1992	Q4	9.5	929157.7217
1993	Q1	9.5	816265.7303
1993	Q2	9.5	903750.2602
1993	Q3	7.75	880279.1142
1993	Q4	8.25	820364.5508
1994	Q1	6.25	943853.7394
1994	Q2	5.5	820536.3950
1994	Q3	5	983587.8272
1994	Q4	5	982711.0565
1995	Q1	5	892617.0649
1995	Q2	6	842623.9421
1995	Q3	6	865700.4039
1995	Q4	5	833365.6841
1996	Q1	4.25	873456.9240
1996	Q2	3.75	971276.6635
1996	Q3	3.25	888702.0169
1996	Q4	3.25	838152.3805
1997	Q1	3.25	928228.6676
1997	Q2	3.25	956261.3857
1997	Q3	3.25	1012012.9409
1997	Q4	3.25	1058916.4467
1998	Q1	3.5	1046895.8280

Figure 12: Nominal interest rate vs. average purchase price (1992–1998 shown) — early 1990s show high interest rates (9.5%) with relatively low property prices, confirming inverse relationship

The data clearly shows that as nominal interest rates fell from 9.5% in 1992 toward 3.25% by the late 1990s, average purchase prices rose from around DKK 820K to DKK 1.05M. This inverse relationship holds throughout the full dataset, with the lowest rates in 2020–21 coinciding with price peaks.

Q11 — Real Interest Rate vs. SQM Price

Year	Quarter	real_interest_rate	avg_sqm_price
2024	Q4	1.2800000000000007	22159.915651948046
2024	Q3	2.2199999999999971	22476.875865508675
2024	Q2	2.4699999999999919	23815.768867084134
2024	Q1	2.4699999999999945	22334.46427920952
2023	Q4	0.29000000000000083	22716.104939247343
2023	Q3	-0.21000000000000024	22523.527279226848
2023	Q2	-0.71000000000000142	21929.84386831746
2023	Q1	-1.5599999999999608	22419.297100329488
2022	Q4	-7.0500000000000119	20148.786539577075
2022	Q3	-7.699999999999989	20516.554592282704
2022	Q2	-7.70000000000001405	22374.060468346255
2022	Q1	-7.7000000000000181	23242.355798016823
2021	Q4	-1.8499999999999464	22734.933503337517
2021	Q3	-1.849999999999943	22017.009639310323
2021	Q2	-1.8499999999999417	21400.828946191545
2021	Q1	-1.849999999999945	21619.194719399053
2020	Q4	-0.41999999999998805	21497.654264160657
2020	Q3	-0.41999999999998944	20690.778565731278
2020	Q2	-0.41999999999999177	19194.014589430877

Figure 13: Real interest rate vs. average SQM price by quarter (2020–2024) — negative real rates in 2022 coincide with peak SQM prices; as real rates turned positive in 2023, SQM growth moderated

Interest Rate Impact Insights

- A clear inverse relationship exists between interest rates and property prices — lower rates accelerate price growth, higher rates compress it.
- The 2022 period of sharply negative real interest rates (nominal rate < inflation) was a major driver of the 2021-22 price surge.
- As real interest rates turned positive in 2023-24 (reaching 1.28% by Q4 2024), SQM price growth plateaued at DKK ~22,000–22,500/m².
- This relationship is critical for investment timing: periods of low-to-negative real rates are historically associated with price acceleration.

Q12 — Inflation Regimes vs. Property Prices

The SQL analysis categorized transactions into three inflation regimes. High inflation periods (above 3%) are associated with strong purchase price levels as property is used as an inflation hedge. However, analysis is partial as the output covers a subset of data — full conclusions require the complete dataset view.

5.5 Negotiation & Pricing Gap Analysis

Q13 — Offer-to-Purchase Discount by Region

Average offer-to-purchase discount percentages vary by region. The Power BI dashboard shows Zealand buyers negotiate the largest absolute gap (avg -DKK 55K), followed by Fyn & Islands (-DKK 39K), Jutland (-DKK 34K), and Bornholm (-DKK 19K). Higher-value markets see larger absolute negotiation gaps.

Q14 — Cities with Highest Negotiation Gap

The top 5 cities with the most extreme negotiation gaps represent high-value markets where the spread between asking and final price is largest. These cities likely have longer days-on-market and more motivated sellers, creating opportunities for buyers.

Q15 — Negotiation Gap by Price Segment

Premium and Ultra-Luxury segments exhibit larger average negotiation gaps, consistent with the general principle that higher-priced assets have more pricing flexibility. Budget segment buyers face near-zero negotiation gaps, indicating tight competition in entry-level markets.

5.6 Advanced Analysis: Growth & Value Screening

Q16 — Year-over-Year City Price Growth

city	Year	current_year_avg	next_year_avg	percent_price_change_to_next_year
Aalborg	2015	2185627.2727	2302515.4167	5.34803649
Aalborg	2018	2302515.4167	2580548.3492	12.07518223
Aalborg	2019	2580548.3492	2522730.8250	-2.24051311
Aalborg	2020	2522730.8250	2651577.0857	5.10741215
Aalborg	2021	2651577.0857	2505214.2857	-5.51983952
Aalborg	2022	2505214.2857	2387334.9365	-4.70535993
Aalborg	2023	2387334.9365	NULL	NULL
Aarhus C	2016	3209078.0862	3598662.8548	12.14008379
Aarhus C	2018	3598662.8548	3244248.2571	-9.84850796
Aarhus C	2019	3244248.2571	3129811.4375	-3.52737554
Aarhus C	2020	3129811.4375	3678423.3243	17.52859231
Aarhus C	2021	3678423.3243	3888200.0000	5.70289652
Aarhus C	2022	3888200.0000	3339141.6351	-14.12114513
Aarhus C	2023	3339141.6351	NULL	NULL
Ebeltoft	2021	1905070.3704	NULL	NULL
Fredericia	2018	2429614.6545	2072967.8571	-14.67915073
Fredericia	2020	2072967.8571	2655234.5000	28.08855144
Fredericia	2021	2655234.5000	NULL	NULL
Frederik...	2019	4206961.8793	4775519.0909	13.51467467
Frederik...	2021	4775519.0909	3819585.9375	-20.01736639
Frederik...	2023	3819585.9375	NULL	NULL
Haderslev	2021	1843105.6271	NULL	NULL
Herning	2019	1716523.2881	2075462.3333	20.91081710
Herning	2020	2075462.3333	2147286.8194	3.46064995
Herning	2021	2147286.8194	NULL	NULL

Figure 14: Year-over-year price growth by city (partial view) — Aarhus C shows 12.1% growth in 2016 and 17.5% in 2020; Aalborg shows 12.1% growth in 2018; some years show negative growth indicating market corrections

□ YOY Growth Insights

- Aarhus C shows strong YOY growth (12–17%) in multiple years, positioning it as a high-growth secondary market worth monitoring.
- Aalborg shows consistent positive growth with occasional dips, typical of a healthy but cyclical market.

- Frederiksberg shows a sharp reversal in 2021 (-20%) after a peak, highlighting the importance of entry timing in high-value markets.
- NULL next-year values represent the most recent year per city where forward comparison is unavailable — not data errors.

Q17 — Properties Sold Above 20% Premium vs. City Average

house_id	city	sqm_price	city_avg_sqm_price
1312465	Åbyhøj	74568.97	23510.321604191606
89944	Åbyhøj	66666.664	23510.321604191606
1376441	Åbyhøj	56567.164	23510.321604191606
1460633	Åbyhøj	50885.715	23510.321604191606
1376463	Åbyhøj	49868.42	23510.321604191606
234542	Åbyhøj	48809.523	23510.321604191606
693181	Åbyhøj	48057.69	23510.321604191606
70400	Åbyhøj	47834.645	23510.321604191606
96620	Åbyhøj	46250	23510.321604191606
104967	Åbyhøj	45185.184	23510.321604191606
472669	Åbyhøj	44525.547	23510.321604191606
473568	Åbyhøj	43571.43	23510.321604191606
93121	Åbyhøj	42962.965	23510.321604191606
79175	Åbyhøj	42000	23510.321604191606
33610	Åbyhøj	41200	23510.321604191606
217515	Åbyhøj	39534.883	23510.321604191606
1080437	Åbyhøj	38274.336	23510.321604191606
109367	Åbyhøj	36824.324	23510.321604191606
437804	Åbyhøj	36770.074	23510.321604191606
5661	Åbyhøj	36690.65	23510.321604191606
377911	Åbyhøj	36637.93	23510.321604191606
668082	Åbyhøj	36320.754	23510.321604191606
191496	Åbyhøj	36250	23510.321604191606
721465	Åbyhøj	36111.11	23510.321604191606
478077	Åbyhøj	36111.11	23510.321604191606

Figure 15: Properties with SQM price at least 20% above city average (partial view, Åbyhøj city shown) — individual house_ids with SQM prices ranging DKK 36K–74K vs city average of DKK 23,510/m²

This window function query identifies premium-performing properties — those commanding at least a 20% SQM price premium over their city's average. These represent either genuinely superior properties (location, condition, size) or outliers that merit investigation. In Åbyhøj, the city average SQM price is DKK 23,510 while flagged properties range from DKK 36,111 to DKK 74,569 — showing the extent of premium pricing within the same city.

Q18 — Best Value House Type

This composite query evaluates all house types by average price, SQM price, room count, and price-per-room. Summerhouses offer the lowest entry price (DKK 1.22M avg) with the lowest SQM price, making them accessible. Villas dominate by volume and offer the best price-per-room value relative to their total size, while Apartments offer premium location access at a higher SQM cost.

6. Power BI Dashboard

An interactive 3-page Power BI dashboard was developed to present the full analysis visually. The dashboard includes cross-filtering capabilities with slicers for Year, City, Region, and House Type across all pages.

6.1 Page 1: Market Overview

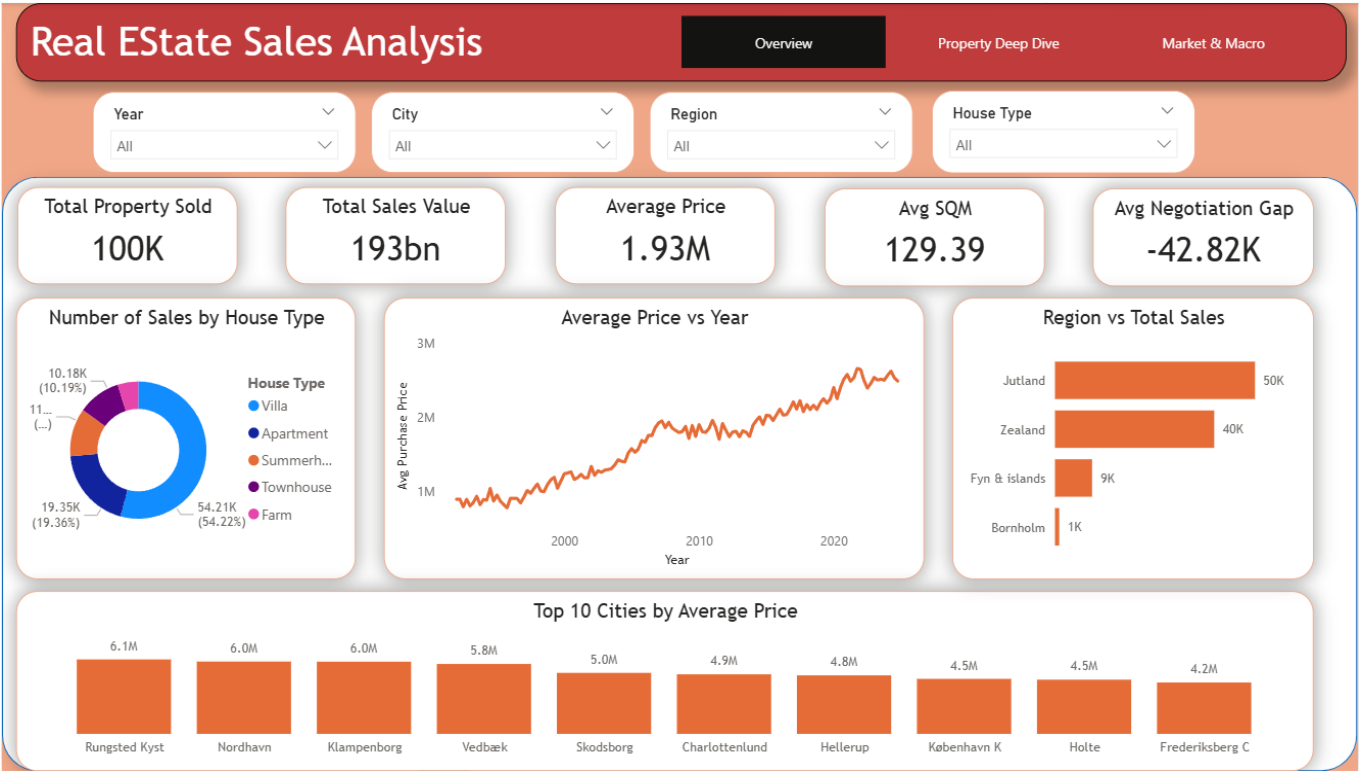


Figure 16: Dashboard Page 1 — Market Overview showing KPIs, house type mix, price trend (1992–2024), regional volume, and Top 10 cities by average price

Dashboard Page 1 — Key Findings

- Total market: 100K transactions worth DKK 193Bn at an average price of DKK 1.93M with average area of 129 m².
- Villas dominate at 54.22% of all transactions; Apartments are second at 19.36%.
- Average price trend shows consistent growth from ~DKK 800K in 1992 to ~DKK 2.8M in 2024, with a visible plateau in 2022–23.
- Jutland leads in transaction volume (50K) but Zealand is the premium market by price.
- Rungsted Kyst, Nordhavn, and Klampenborg are the top 3 most expensive cities — all in the Copenhagen metropolitan North Shore area.
- Average negotiation gap stands at -DKK 42.82K — sellers accept offers below list in the aggregate, confirming active buyer negotiation.

6.2 Page 2: Property Deep Dive

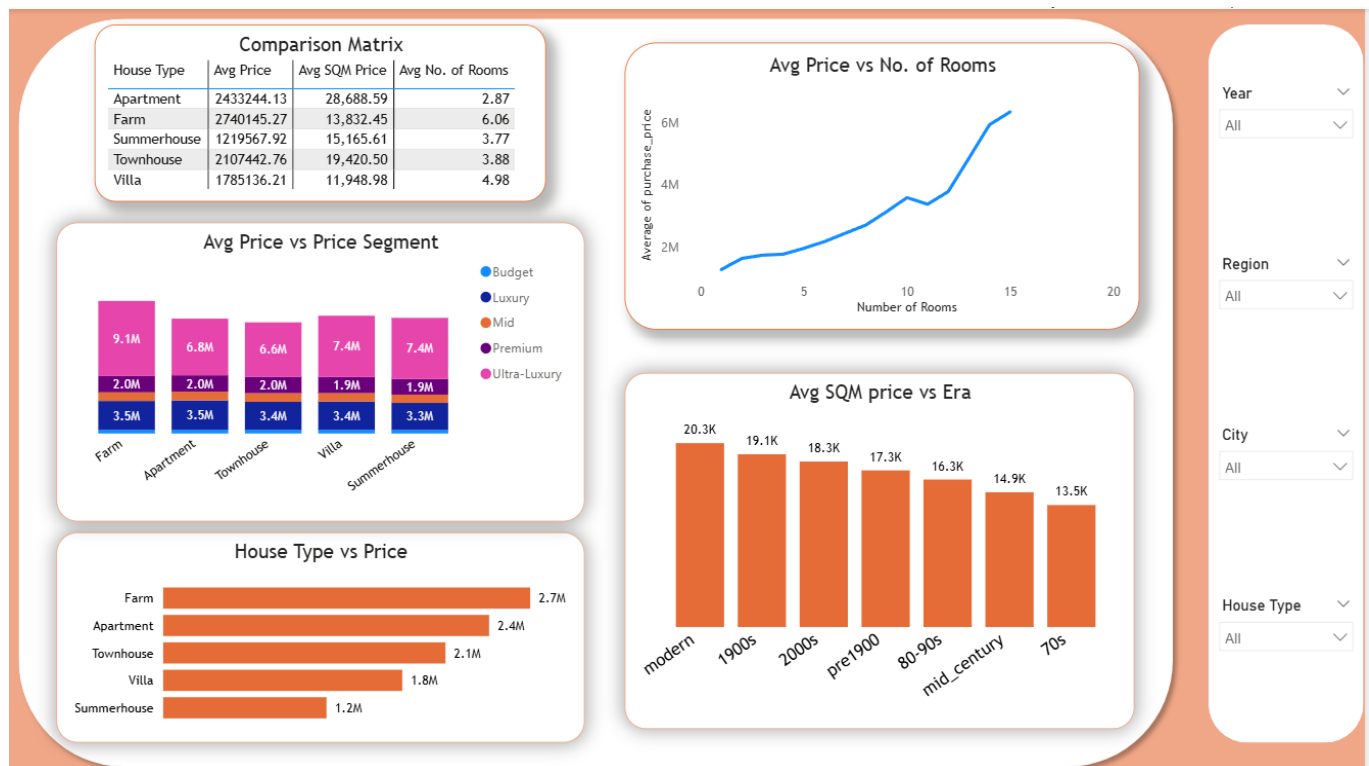


Figure 17: Dashboard Page 2 — Property Deep Dive showing comparison matrix, price by segment, rooms-to-price relationship, and SQM price by build era

Dashboard Page 2 — Key Findings

- Farm has the highest average purchase price (DKK 2.74M) and the most rooms (6.06 avg), but the lowest SQM price (DKK 13,832) — size-driven, not location-driven.
- Apartment leads in SQM price (DKK 28,689) with fewest rooms (2.87 avg) — compact urban premium assets.
- Modern-era properties command DKK 20.3K/m² while 70s-era properties fetch only DKK 13.5K/m² — a 50% premium for contemporary construction.
- Ultra-Luxury segment exists across all house types with the highest concentration in Farms (DKK 9.1M avg in Ultra-Luxury tier).
- A positive near-linear relationship exists between number of rooms and average price, peaking above DKK 6M for 15+ room properties.

6.3 Page 3: Market & Macro

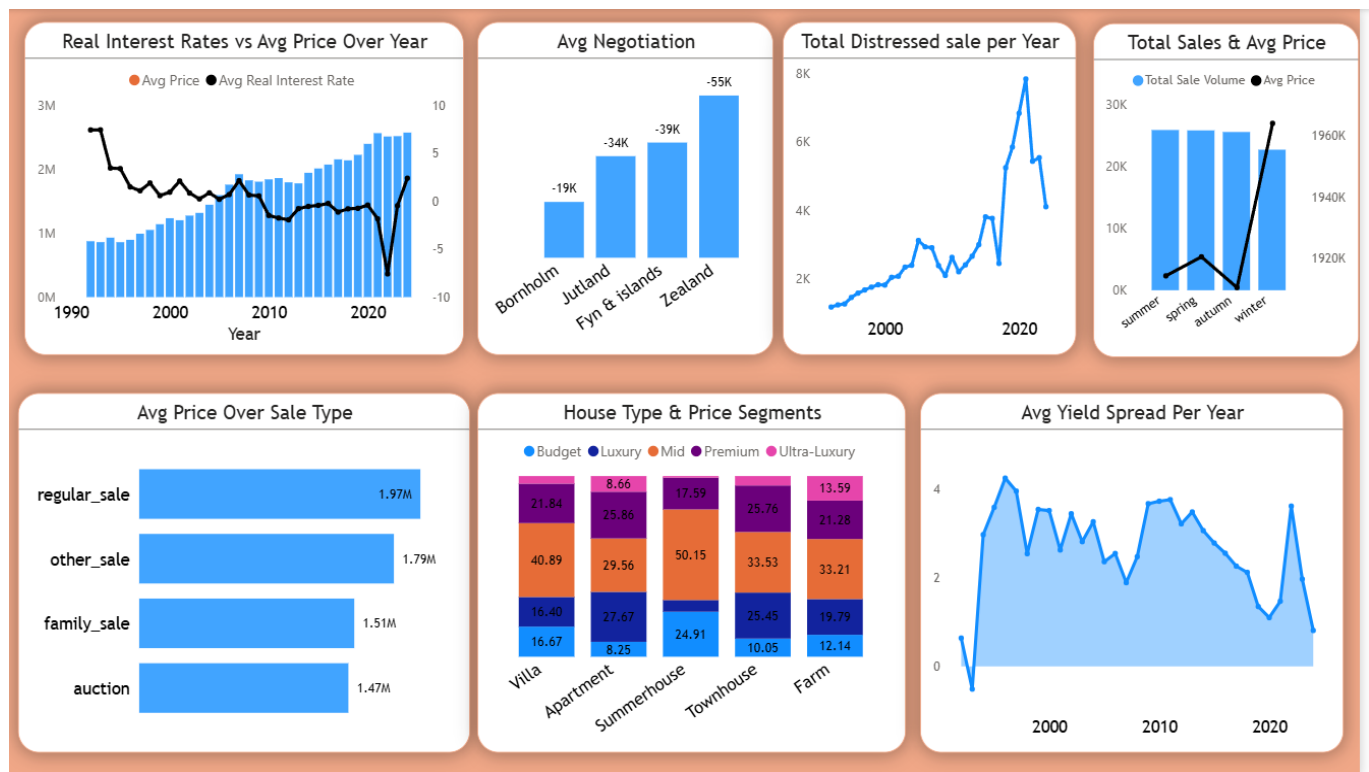


Figure 18: Dashboard Page 3 — Market & Macro showing real interest rates vs price, negotiation by region, distressed sales, seasonal patterns, sale type analysis, and mortgage yield trend

Dashboard Page 3 — Key Findings

- Real interest rate and property price show a clear inverse relationship over 30+ years — as real rates fell from ~8% in the early 1990s toward negative territory in 2021, prices more than tripled.
- The sharp negative real interest rate spike in 2022 (approx. -8%) coincided with peak property prices before rate normalization.
- Zealand buyers negotiate the largest absolute gap (-DKK 55K avg), while Bornholm has the smallest (-DKK 19K) — reflecting market competitiveness.
- Total distressed sales peaked sharply around 2007–2009 (financial crisis) and again showed a smaller uptick post-2020. Overall distress rates remain low (2–4%).
- Winter sales have the highest average price (DKK ~1,960K) while Summer has the lowest volume — buyers in winter may be more motivated and purchase higher-value properties.
- Regular sales average DKK 1.97M vs. Auction sales at DKK 1.47M — auctions present a 25% discount opportunity.
- Mortgage bond yield (Avg Yield Spread) peaked ~4% in the mid-2000s and trended toward near-zero in 2020, before rising again post-2022 — tracking the broader interest rate cycle.

7. Summary of Key Findings

#	Key Finding
1	Denmark's residential market grew from DKK ~820K average (1992) to DKK ~2.8M (2024) — a ~240% increase over 32 years.
2	Villas account for 54% of all transactions, making them the core product of the Danish housing market.
3	Apartments command the highest SQM price (DKK 25,437–28,689/m²), driven by urban location premiums in Copenhagen-area cities.
4	North Zealand coastal cities (Rungsted Kyst, Klampenborg, Nordhavn) are the highest-value markets averaging DKK 5.8M–6.1M.
5	Modern properties (post-2010) sell at a 50%+ SQM premium over 70s-era builds, reflecting energy efficiency and modern standards.
6	Real interest rates have the strongest macro influence on prices — negative real rates (2020–22) corresponded to peak price appreciation.
7	2021 Q1 was the peak activity quarter with 2,263 transactions and DKK 5.9Bn in total sales value.
8	64% of all transactions have zero negotiation gap — when negotiation occurs, Farms see the deepest discounts (avg -DKK 237K).
9	Zealand has the lowest distress rate (2.10%) confirming it as the most liquid and stable market; Bornholm has the highest (3.97%).
10	Auction properties trade at a ~25% discount to regular sales, representing a systematic value entry point for investors.
11	Cities like Aarhus C show consistent 12–17% YOY price growth in multiple years, qualifying as a high-growth secondary market.
12	Summerhouses are the most negotiated segment (40.4% see a price drop), but average discount (-DKK 94K) is the smallest.

8. Business Recommendations

Based on the full analysis across Python EDA, SQL business queries, and Power BI dashboard insights, the following strategic recommendations are made for a real estate consulting and investment firm operating in the Danish market.

8.1 Investment Strategy Recommendations

Recommendation 1: Prioritize Zealand for Core Investment

Zealand — particularly the North Shore corridor (Rungsted Kyst, Klampenborg, Vedbæk) — offers the highest property values, the lowest distressed sale rate (2.10%), and the broadest price IQR spread (DKK 2.1M). This market exhibits the strongest demand fundamentals and should be the primary focus for premium portfolio allocation.

Recommendation 2: Target Aarhus C and Aalborg as Growth Markets

Aarhus C shows 12–17% YOY price growth in multiple years, making it the strongest secondary growth market identified. Aalborg shows consistent positive growth. Both offer entry price points significantly below Copenhagen-area premiums while delivering superior capital appreciation rates. A buy-and-hold strategy in Aarhus C Apartments or modern-era Townhouses is well-supported by the data.

Recommendation 3: Use Auctions as a Value Entry Strategy

Auction properties sell at an average 25% discount versus regular sales (DKK 1.47M vs DKK 1.97M). Investors with capacity to assess and renovate distressed properties should actively monitor auction pipelines — particularly in Bornholm and Fyn where distress rates are higher — to capture forced-sale discounts.

Recommendation 4: Modern and 2000s Build Era Properties for Premium Returns

Modern-era (post-2010) properties command a 50%+ SQM premium over 70s and 80-90s builds. Investors should avoid 70s-80s era properties unless the acquisition price reflects the renovation discount. New-build acquisition or conversion investment in the Modern tier offers the best price-per-m² growth trajectory.

8.2 Pricing & Product Strategy

Recommendation 5: Use SQM Price as the Core Valuation Metric

Total purchase price varies enormously by property size, making comparisons misleading. SQM price provides a standardized, like-for-like valuation basis. The firm's pricing models and client advisory tools should default to SQM price as the primary comparator, especially for cross-city and cross-type analysis.

Recommendation 6: Exploit the Room Count Premium

There is a near-linear positive relationship between room count and average purchase price. For development or acquisition decisions, properties with 8+ rooms are systematically underrepresented in supply but show strong average prices. Larger family-oriented villas and townhouses in mid-market cities represent an under-served segment.

Recommendation 7: Differentiate Negotiation Strategies by Property Type

Farm and Villa buyers should expect to negotiate (39–40% of transactions see a gap) and can target meaningful discounts (DKK 125K–237K on average). Apartment buyers face stiffer pricing — urban demand keeps sellers firm (only 31% see negotiation). Summerhouse buyers should negotiate consistently but expect modest gains. Pricing advisory to clients should be calibrated accordingly.

8.3 Macro-Timing & Risk Management

Recommendation 8: Monitor Real Interest Rate as the Primary Market Timing Signal

The analysis confirms that real interest rate (nominal rate minus inflation) is the most predictive macro variable for Danish property price direction. When real rates are negative or near-zero, the market is in an expansion phase — accelerate acquisitions. When real rates rise above 2–3%, expect price moderation or correction — reduce exposure or target value plays.

Recommendation 9: Avoid Over-Exposure to Bornholm

Bornholm's 3.97% distressed sale rate and small market size signal weaker demand and lower liquidity. While entry prices may appear attractive, exit risk is elevated. The firm should limit Bornholm exposure to opportunistic, short-hold strategies rather than core portfolio allocation.

Recommendation 10: Leverage the Winter Buying Window

Winter sees the highest average purchase prices (DKK 1,960K+) but also the lowest volume — suggesting that winter buyers are motivated, well-qualified purchasers of higher-value properties. Sellers listing in winter may transact faster. The firm should time premium property launches for autumn and track winter closing volumes as a demand health indicator.

9. Technology Stack & Tools

Tool	Purpose	Key Libraries / Features Used
Python 3.x	Data Cleaning & EDA	Pandas, NumPy, Matplotlib, Seaborn, SQLAlchemy
Jupyter Notebook	Interactive Analysis	Cell-based exploration, inline visualizations
MySQL	SQL Business Analysis	Window functions (LEAD, OVER PARTITION BY), GROUP BY aggregations, subqueries, CASE statements, CTEs
Power BI Desktop	Dashboard & Visualization	Custom DAX measures, cross-page filtering, slicers, interactive charts
GitHub	Version Control	Project repository with ipynb, SQL, and documentation

Project Pipeline

Business Problem Definition	Data Cleaning & EDA (Python)	SQL Analysis 18 Queries (MySQL)	Power BI 3-Page Dashboard	Project Report Insights & Recs	GitHub Repo Code & Portfolio
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About This Project

This is an end-to-end data analytics portfolio project covering the full analytical pipeline from business problem definition to dashboard delivery. The analysis covers 100,000 Danish residential property transactions spanning 32 years, combining Python-based data science, SQL-driven business querying, Power BI visualization, and structured strategic reporting. All code and queries are available in the accompanying GitHub repository.